

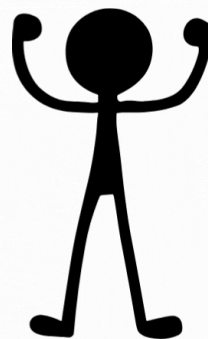
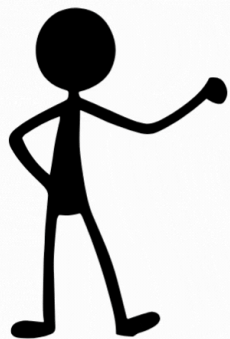
Signal-Aware Framework for Multilingual LM Evaluation

María Grandury, Angelika Romanou, Éléonore Hasler,

Clara Meister, Antoine Bosselut

EPFL NLP

Every now and then...



Agenda

1 Introduction

2 Related Work

3 Methodology

4 Analysis

Introduction

Training multilingual LMs requires constant evaluation decisions,
but evaluation is expensive and often uninformative.

Training LMs has a **high cost** and requires **constant decisions** (data mixtures, hyperparameters, etc.) guided by benchmark evaluations. However,

Not all benchmarks provide informative signals

- ⚠ **High variance:** scores fluctuate across runs
- ⚠ **Redundancy:** multiple benchmarks measure the same thing
- ⚠ **Weak correlation:** improvements don't reflect real progress
- ⚠ **Cost:** large suites are expensive to run frequently

This assumption is **especially fragile** in multilingual settings

- 🌐 Low-resource languages are underrepresented in training data
- 🌐 Small models may perform **near-randomly** on harder tasks
- 🌐 Linguistic diversity introduces new sources of variability
- 🌐 Existing reliability tools were validated **only on English**



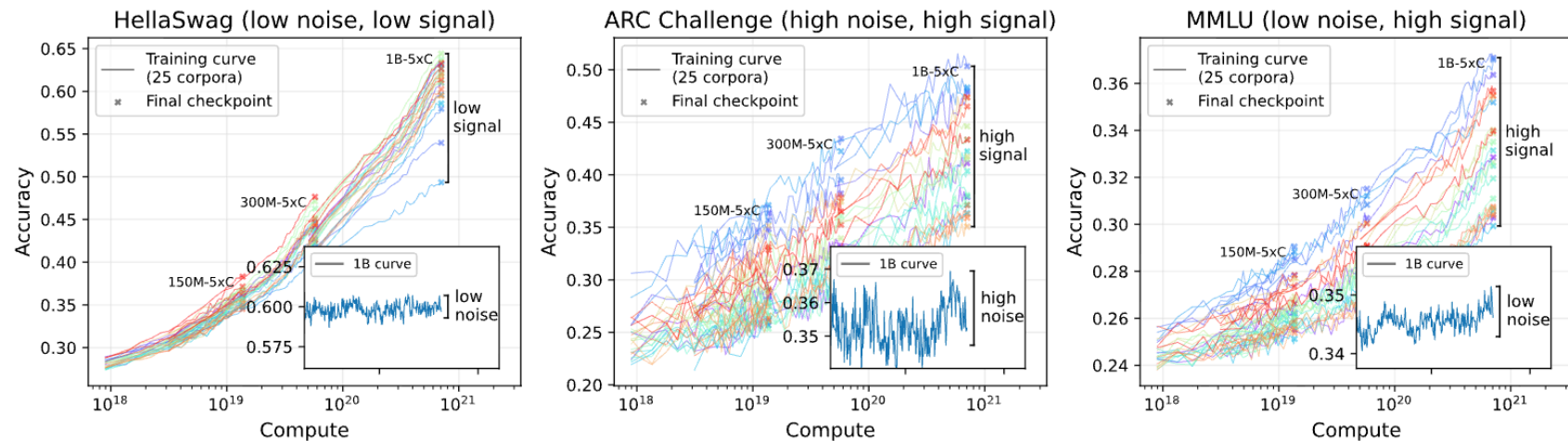
Which (subsets of) benchmarks provide reliable signal at each stage of multilingual model training?

Related Work

Insights from the Signal-and-Noise Paper, Heineman et al. (2025), AllenAI

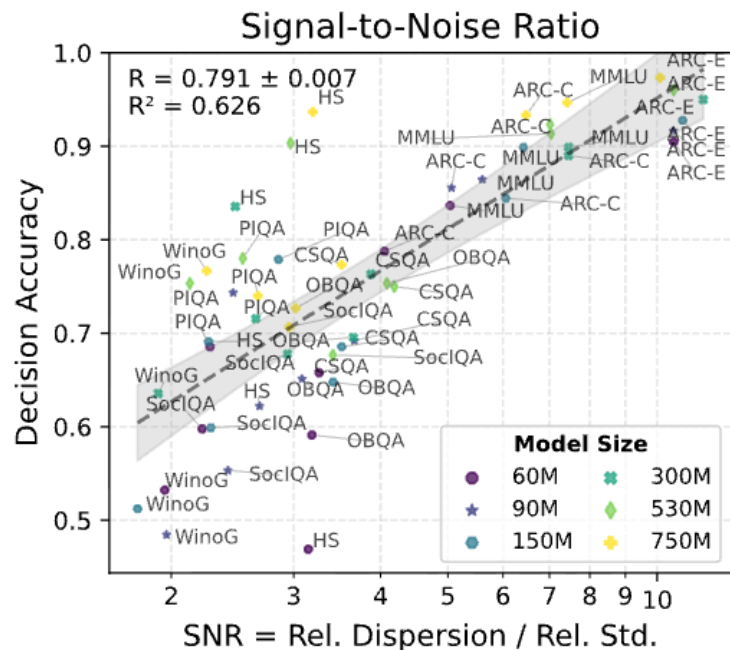
Signal-and-Noise Paper

Figure 1. Examples of signal and noise



Signal-and-Noise Paper

Figure 2. Correlation between SNR and Decision Accuracy



- **SNR predicts decision accuracy** ($R = 0.791$), while signal or noise alone do not
- **Noise predicts scaling-law error** ($R = 0.653$): noisier benchmarks have less reliable extrapolation
- **Filtering subtasks by SNR** yields subsets that outperform full benchmarks (e.g., 16/57 MMLU subtasks $\rightarrow$ +2.6% DA)
- **Averaging checkpoint scores** reduces noise and improves decision accuracy (+2.4% on 30-task avg.)
- **Bits-per-byte (BPB)** improves DA over accuracy on 90% of benchmarks, especially math and code

Methodology

The Signal-Aware Framework, multilingual edition

Decision Accuracy

For all pairs of small models (s_a, s_b) trained on datasets (a, b) and their large versions (m_a, m_b) , does the ranking for task B hold?

$$\text{DA} = \frac{1}{|\mathcal{P}|} \sum_{(a,b) \in \mathcal{P}} 1[\text{sign}(B(s_a) - B(s_b)) = \text{sign}(B(m_a) - B(m_b))]$$

Pretraining:

- **DA-size:** small-size $\rightarrow$ large-size rank agreement (cross-scale)
- **DA-ckpt:** within-size early $\rightarrow$ late checkpoint rank agreement

Posttraining (additional):

- **DA-stage:** cross-posttraining stage (SFT, DPO, RLVR)
- **DA-ctx:** smaller-larger context rank agreement



Decision Accuracy is what we ultimately
want a benchmark to get right
We will look for other metrics that are **proxies** we can
compute cheaply during training.

Signal (Relative Dispersion / Normalized Max. Difference)

How well a benchmark b separates a pair of model scores m_j, m_k of similar scale s trained on different settings:

$$\text{Rel. Dispersion}(b, s) = \frac{\max_{j,k} |m_j - m_k|}{\bar{m}}$$

Noise (Relative Std. Dev.)

Benchmark b variability across the final n training checkpoints of a model m :

$$\text{Rel. Std.}(b, m) = \frac{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (m_i - \bar{m})^2}}{\bar{m}}$$

$$\text{SNR} = \frac{\text{Avg. Rel. Dispersion}(\text{final checkpoint})}{\text{Rel. Std.}(\text{final } n \text{ checkpoints})}$$

Signal = how much a benchmark separates models. There are **many ways to quantify "spread"**, we consider **22 candidate variants** grouped into 5 mathematical families:

Family	Members	Idea
Dispersion	mpd , aad , rms_deviation , quartile_deviation , dist_std , dispersion , range	Absolute spread of scores
Relative-spread	rel_std , rel_mpd , rel_mpsd , iqr , rel_dispersion	Spread normalized by mean (AllenAI default rel_std)
Discrepancy	discrepancy , star_discrepancy , star_discrepancy_shifted , rel_star_discrepancy , dispersion_shifted , gini	Uniformity / inequality of the score distribution
Robust	mad , mpsd	Outlier-resistant spread
Depth	tukey , projection	Half-space statistical depth

Q1: Which variant best tracks decision accuracy across languages?

Each variant is an alternative **signal** numerator (SNR = signal / noise). Scores c_i = per-mix final accuracies at one size; mean $\bar{c}$ (μ), std σ , quartiles Q_1, Q_3 , empirical CDF F_n .

Dispersion · absolute spread

dispersion	$\max_{i,j} c_i - c_j $
range	$\max c - \min c$
mpd (mean pairwise dist.)	$\frac{1}{n^2} \sum_{i,j} c_i - c_j $
aad (avg abs deviation)	$\frac{1}{n} \sum_i c_i - \bar{c} $
rms_deviation	$\sqrt{\frac{1}{n} \sum_i (c_i - \bar{c})^2}$
quartile_deviation	$(Q_3 - Q_1)/2$
dist_std	$\text{std}\{ c_i - c_j \}$

Relative-spread · ÷ mean

rel_std	σ/μ
rel_dispersion	$\max_{i,j} c_i - c_j /\bar{c}$
rel_mpd	$\frac{1}{n^2} \sum_{i,j} c_i - c_j /\bar{c}$
rel_mpsd	$\frac{1}{n^2} \sum_{i,j} (c_i - c_j)^2/\bar{c}^2$
iqr (interquartile range)	$(Q_3 - Q_1)/\bar{c}$

Discrepancy · uniformity / inequality

discrepancy	$\max_c F_n(c) - F(c) $
star_discrepancy	$\sup_{[0,c]} F_n - F $
star_discrepancy_shifted	$\sup_{[0,c]} F_n - F $, scaled
rel_star_discrepancy	$\sup_{[0,c]} F_n - F /F$
dispersion_shifted	$\max_{i,j} c_i - c_j $, scaled
gini	$\frac{1}{2n^2\mu} \sum_{i,j} c_i - c_j $

Robust · outlier-resistant

mad (median abs deviation)	$\text{med} c_i - \text{med}(c) $
mpsd (mean pairwise squared dev)	$\frac{1}{n^2} \sum_{i,j} (c_i - c_j)^2$

Depth · half-space

tukey	$\min(F_n(x), 1 - F_n(x))$
projection	$(1 + x - \text{med}(c) /\text{MAD})^{-1}$

ORIGINAL

Checkpoint noise

Std. dev. over **late training checkpoints**, aggregated across models, normalized by benchmark-level mean

$$\text{Noise} = \frac{\frac{1}{|M|} \sum_m \sigma_{\text{step}}(m)}{\mu(M)}$$

⚠ Requires intermediate checkpoints (rarely available)

vs

ÉLÉONORE & CLARA

Benchmark noise

Relative std. dev. across ***k*-fold splits** of the evaluation set, averaged across models

$$\text{Noise} = \frac{1}{|M|} \sum_m \frac{\sqrt{\frac{1}{k} \sum_{i=1}^k (m_i - \bar{m})^2}}{\bar{m}}$$

- ✓ Computable from a **single evaluation run**
- ✓ More predictive of decision accuracy in preliminary results from Eléonore and Clara.

The right benchmark depends on **where in training** you are.

Stage	Decision being made	Models available
Pretraining	Data mixture, architecture, scale	Custom 175M-1B suite (controlled)
Midtraining	Domain/quality up-sampling	Open-source bases (3B-70B)
Post-training	SFT / DPO / RLVR recipe	Instruct families (3B-70B)

- We compute signal, noise, SNR, DA and scaling-law error **per stage**
- Goal: **stage-specific evaluation recommendations**, not a single global ranking
- Current results focus on the **pretraining** stage (the controlled custom suite)

Experimental Setup

Models and benchmarks

Experimental Setup – Models

Controlled custom pretraining suite (36 models)

36 Apertus pretrains = 4 sizes × 3 data mixtures × 3 seeds, each trained on **100B tokens** (50k iterations).

Axis		Values					
Sizes		175M, 350M, 600M, 1B					
Mixtures (FineWeb-Edu / FineWeb2)		30 / 70, 60 / 40, 90 / 10					
Seeds		28, 1797, 1904					

Label	Layers	d_model	Head dim	Heads	KV Heads	FFW Mult	Non-emb Params
175M	16	1024	64	16	4	4	0.176B
350M	20	1280	64	20	5	4	0.344B
600M	24	1536	64	24	6	4	0.595B
1B	28	1792	64	28	7	4	0.944B

Experimental Setup – Data Mixtures

FineWeb-Edu (EN) + FineWeb2 (multilingual)

Data Mixtures

- **English:** FineWeb-Edu (DCLM)
- **Multilingual:** FineWeb2
 - Apertus' high-quality filter
 - Top 200 languages
 - Original naturally-occurring language distribution
- Mixtures: **30/70, 60/40, 90/10**
- Tokens: **100B** in total

Differences with Apertus: tied embeddings (128k vocab), no goldfish loss, no cross-document attention masking.

Language	Documents	Percentage (%)
Russian (rus.Cyrl)	605,468,615	13.26%
Mandarin Chinese (cmn.Hani)	578,332,129	12.66%
German (deu.Latn)	427,700,394	9.36%
Spanish (spa.Latn)	405,634,303	8.88%
Japanese (jpn.Jpan)	376,134,745	8.23%
French (fra.Latn)	332,646,715	7.28%
Italian (ita.Latn)	219,117,921	4.80%
Portuguese (por.Latn)	189,851,449	4.16%
Polish (pol.Latn)	138,337,436	3.03%
Dutch (nld.Latn)	133,855,612	2.93%
Indonesian (ind.Latn)	92,992,647	2.04%
Turkish (tur.Latn)	88,769,907	1.94%
Czech (ces.Latn)	62,703,458	1.37%
Korean (kor.Hang)	58,160,164	1.27%
Standard Arabic (arb.Arab)	57,752,149	1.26%
Romanian (ron.Latn)	54,128,784	1.19%
Persian (fas.Arab)	51,043,666	1.12%
Ukrainian (ukr.Cyrl)	47,552,562	1.04%
Hungarian (hun.Latn)	46,879,826	1.03%
Swedish (swe.Latn)	45,329,979	0.99%
Modern Greek (1453-) (ell.Grek)	44,202,550	0.97%
Danish (dan.Latn)	42,975,661	0.94%
Vietnamese (vie.Latn)	40,741,340	0.89%
Thai (tha.Thai)	35,949,449	0.79%
Norwegian Bokmål (nob.Latn)	35,502,989	0.78%
Finnish (fin.Latn)	33,162,591	0.73%

Experimental Setup – Models

Scaling beyond 1B: open-source families

The custom suite caps at 1B. Open-source families extend the compute axis to **70B** and cover the mid-/post-training stages.

Size	Pretraining	Midtraining	Post-training
$\leq 1\text{B}$	Custom 175M–1B (3 mixes $\times$ 3 seeds); Apertus3 1B; distilled 0.6B/1B	–	distilled Apertus 1.7B SFT
3B	Apertus3 3B	SmolLM3-3B base	SmolLM3-3B
7-9B	–	Apertus-8B base; OLMo-3-7B base	Apertus-8B Instruct; OLMo-3-7B SFT/DPO/Instruct
$\geq 12\text{B}$	–	gemma-3 12-27B; OLMo-3 13-32B; Qwen3 14B+	Apertus 70B Instruct

Experimental Setup – Models

Families spanning ≥ 2 buckets

Families spanning ≥ 2 buckets: Apertus-2509 ['7-9B', '70B'] OLMo-2-base ['1B', '7-9B', '12-14B', '27-32B'] Olmo-3-base ['7-9B', '27-32B'] Qwen3-Base ['600M', '1.7B', '4B', '7-9B', '12-14B'] ap-from8b-TOP256 ['600M', '1B'] apertus-fwEdu30-fw270-seed1797 ['175M', '350M', '600M', '1B'] apertus-fwEdu30-fw270-seed1904 ['175M', '350M', '600M', '1B'] apertus-fwEdu30-fw270-seed28 ['175M', '350M', '600M', '1B'] apertus-fwEdu60-fw240-seed1797 ['175M', '350M', '600M', '1B'] apertus-fwEdu60-fw240-seed1904 ['175M', '350M', '600M', '1B'] apertus-fwEdu60-fw240-seed28 ['175M', '350M', '600M', '1B'] apertus-fwEdu90-fw210-seed1797 ['175M', '350M', '600M', '1B'] apertus-fwEdu90-fw210-seed1904 ['175M', '350M', '600M', '1B'] apertus-fwEdu90-fw210-seed28 ['175M', '350M', '600M', '1B'] apertus3-a06 ['1B', '3B'] gemma-3-pt ['270M', '1B', '4B', '12-14B', '27-32B']

Buckets present: ['175M', '270M', '350M', '600M', '1B', '1.7B', '3B', '4B', '7-9B', '12-14B', '27-32B', '70B']

Experimental Setup – Benchmarks

Pretraining suite: 22 benchmark families, 12+ languages

Category	Pretraining (base suite)	+ Midtraining	+ Post-training
LM & Completion	hellaswag, piqa, global_piqa, xstorycloze, xwinograd, xcopa, xnli, paws, multiblimp	–	–
Commonsense & Reasoning	arc, commonsense_qa, openbookqa, winogrande	mgsm_direct	bbh, drop, gsm8k_cot, hendrycks_math, mathqa, mgsm
Knowledge & QA	mmlu, global_mmlu_full, belebele, triviaqa, squad, include_base_44, agieval, arabic_leaderboard, truthfulqa	–	global_mmlu (gen), blend, cultural_bench, mmlu_flan_cot
Code	–	–	humaneval, mbpp
Instruction & Safety	–	–	ifeval, multi-if, acp_bench, bbq, toxigen, harmbench, aya_redteaming, polyglotoxicity

- **Languages:** en, es, ar, zh, ru, hi, vi, eu, ja, sw, tr, th, te
- **Evaluation:** log-prob 0-shot (pretraining), generative (post-training)

Results

Accuracy vs. compute

Results

Accuracy vs Compute curves

For each benchmark we plot **accuracy vs. compute (FLOPs)**, one training curve per data mixture, across the four model sizes.

 **Signal** = $(\text{max} - \text{min}) / \text{mean}$ of the per-mixture final scores at 1B

 **Noise** = variance of each curve over its late checkpoints

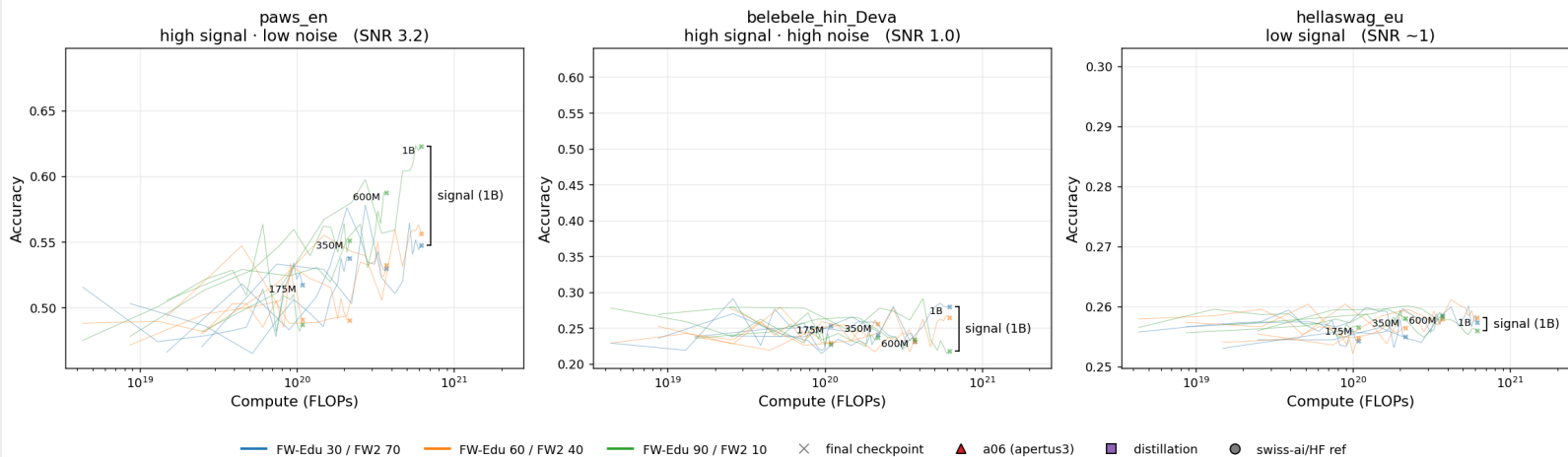
 The benchmarks that **separate the mixtures most** (top Signal at 1B):
`agieval_sat` (0.27), `belebele` (0.24), `arabic_leaderboard`

A benchmark is only useful if its mixtures separate by **more than the noise**.

Results

Accuracy vs Compute curves

Accuracy vs compute — three signal/noise regimes



Results

Accuracy vs Compute (Belebele)

Results

Are the models even above chance?

A benchmark only carries signal if the models clear **chance** ($1 / \text{\#options}$) by a margin. Per (benchmark, size) we average the score over all models; **above random** iff $\text{mean} > \text{chance} + 0.05$. Only **44 of 118 benchmarks** clear chance at any size – and it's an **answer-count** effect:

MCQA options	Random chance	Benchmarks above random
2 (completion / minimal pair)	0.50	28 / 42
3 (xnli)	0.33	7 / 11
4 (knowledge MCQA)	0.25	9 / 63
5	0.20	0 / 2

- The only 4-option survivors: hellaswag (contentful completions) and English arc_easy / arc_challenge
- Translated knowledge MCQA sit **at chance**: belebele 0/12, global_mmlu(_full) 0/16, truthfulqa 0/8, arc_<lang> 0/9
- **The gate is enforced**: random (benchmark, size) cells are dropped from SNR and every downstream RQ

Results

Signal $\neq$ Reliability

The top-Signal families (`belebele` , `agieval_sat` , `arabic_leaderboard`) are exactly the ones the **above-random gate removes** – they sit *at chance*, so they never enter the SNR analysis at all.

- ⚠ They swing a lot with the data mixture (**high signal**) **but are also high-noise**, and the models can't even clear chance on them
- ⚠ Raw mixture sensitivity **is not** reliability → high Signal $\neq$ usable SNR
- ⚠ SNR (signal $\div$ noise) instead ranks `multiblimp` / `hellaswag` (low absolute swing, very low noise) at the **top**

Results - Improvement

Pretrain models with fewer languages

- Currently: 200 languages
- **ATLAS**: Adaptive Transfer Scaling Laws for Multilingual Pretraining (arXiv 2510.22037)
 - For similar loss to a 1B model on 100B English tokens, over n languages:
 - Model size = $1\text{B} \times n^{0.243}$
 - Dataset size = $100\text{B} \times n^{0.728}$

Analysis

Four research questions

RQ1 – SNR Definition

Which SNR definition best correlates with decision accuracy across languages? Does it hold across seeds?

RQ2 – Framework Generalization

Do our Apertus-derived SNR rankings transfer to the AllenAI DataDecide corpus on shared benchmarks?

RQ3 – Subsampling

Can subsets of subtasks or individual items give higher SNR than the full benchmark?

RQ4 – Benchmark Creation

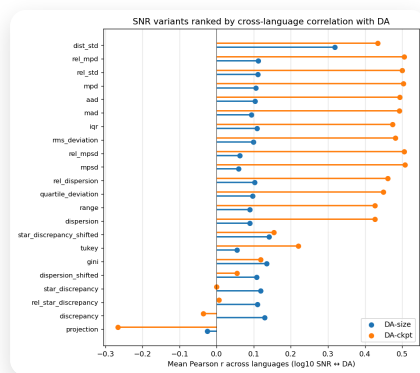
What benchmark design features (curation, format, option count, length) predict high SNR?

RQ1 – SNR Definition

Which SNR variant best predicts decision accuracy? Does it hold across seeds?

RQ1 – SNR Definition

Top variants across languages (DA-size & DA-ckpt)



RQ1 – SNR Definition

Highlighted results

- ✓ **The dispersion family wins.** On `custom_swissai_hf`, `dist_std` is the global best – DA-size r **0.32** (far ahead of the next variant at 0.14), DA-ckpt r **0.43**, overall **0.38**
- ✓ **DA-ckpt is led by the mean-pairwise-distance / relative-spread cluster** (`rel_mpd`, `mpd`, `mpsd` $\approx$ **0.51**) – all dispersion-family members
- ✓ **Better results with more seeds:** top DA-size r climbs **0.31** $\rightarrow$ **0.33** $\rightarrow$ **0.39** (1 $\rightarrow$ 2 $\rightarrow$ 3 seeds)
- ✓ `tukey`, `projection` (depth): **$r \leq 0$** with DA $\rightarrow$ useless at this pool size

RQ1 – SNR Definition

Results (auto) – most reliable benchmark per language (`dist\_std` @ 1B)

lang	top benchmark	SNR	DA-ckpt[?1B]
ar	multiblimp_arb	2.6	0.87
en	xwinograd_en	2.4	0.83
es	multiblimp_spa	3.4	0.85
eu	multiblimp_eus	1.3	0.64
hi	multiblimp_hin	4.9	0.85
ja	xwinograd_jp	2.3	0.76
ru	multiblimp_rus	7.1	0.86
th	xnli_th	1.3	0.75
tr	multiblimp_tur	2.7	0.79
vi	xcopa_vi	1.6	0.76
zh	xcopa_zh	1.6	0.61

RQ1 – SNR Definition

Top-5 benchmarks per language by SNR (dist_std @ 1B)

RQ1 – SNR Definition

Proposed methodology improvements

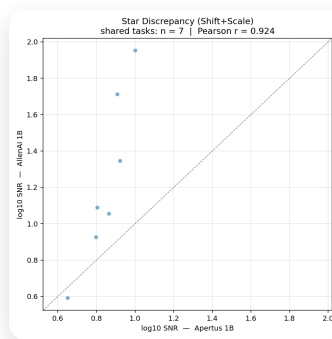
- 💡 **Bootstrap CIs** on per-language Pearson r and cross-pool Spearman ρ
- 💡 Recommend a **family** (dispersion / relative-spread), not an exact variant – only the family transfers
- 💡 Use a **larger DA-size target** (e.g. Apertus-8B) instead of the not-fully-converged 1B custom model

RQ2 – Framework Generalization

Do the SNR variants we recommend correlate with other benchmark reliability frameworks? In particular with AllenAI and FineTasks?

RQ2 – Framework Generalization

Apertus vs AllenAI SNR – pure 3-seed pool, best variant (star_discrepancy_shifted)



RQ2 – Framework Generalization

Highlighted results

- ✓ On the pure 3-seed pool the SNR **values and rank order agree** across corpora over the 7 shared English tasks: **Pearson $r = 0.92$, Spearman $\rho = 0.93$** (best variant `star_discrepancy_shifted`)
- ✓ Value correlation rises with seeds: **0.75** (1 seed) $\rightarrow$ **0.92** (3 seeds)
- ✓ **Dispersion + discrepancy families transfer**; the relative-spread family (incl. AllenAI's own default `rel_std`) does **not**
- ✓ Only **7** English tasks overlap $\rightarrow$ top-K set overlap is uninformative ($K \geq 7 = \text{whole universe} \Rightarrow \text{Jaccard trivially } 1.0$); **the correlation, not the overlap, is the result**

RQ2 – Framework Generalization

Results (auto) – cross-corpus agreement with AllenAI by pool

pool	best variant	Pearson r	Spearman ρ	n_shared
seeds_1904	projection	0.90	0.80	4
seeds_28_1797	dispersion_shifted	1.00	1.00	4
seeds_28_1797_1904	dispersion_shifted	0.98	1.00	4
custom_swissai_hf	mpsd	1.00	1.00	4

Pure pools share all 7 English tasks; `custom_swissai_hf` shares fewer after the above-random gate – the pure 3-seed pool is the like-for-like fit.

RQ2 – Framework Generalization

Proposed improvements

- 💡 **Increase the shared tasks:** add `mmlu` + `mmlupro`, BBH, AGI-Eval
- 💡 **Bootstrap CIs** on the cross-corpus r ($n=7$ is fragile)
- 💡 Compare with other frameworks:
 - **FineTasks**
 - **SMART** filtering (Gupta et al., 2024)
 - How to Select Datapoints for Efficient Human Evaluation of NLG Models? (arXiv 2501.18251)
 - **Chen et al. (2024)** – scaling behavior of downstream tasks
 - **Zhou et al. (2025)** – item response theory for benchmark reliability

RQ3 – Subsampling

Can a subset of subtasks (languages, subjects) or individual items give higher SNR than the full benchmark?

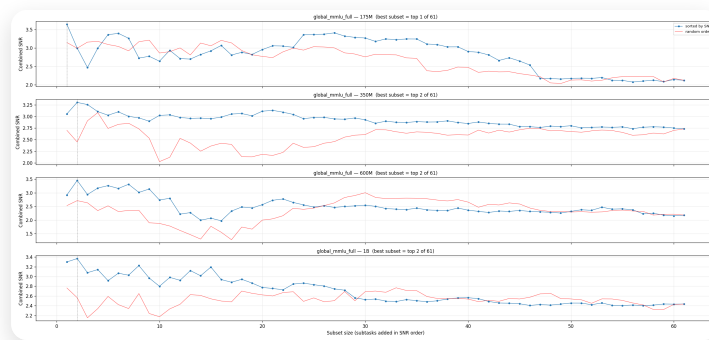
- ⚙ Model pool: 36 Apertus + external pretraining models
(`custom_swissai_hf`), last-5 ckpts per model
- ⚙ **Subtask level**, three cases: language subset, MMLU subject subset, subject × language
- ⚙ **Per-sample level**: rank by sample SNR, remove the ones with signal = 0, add samples greedily, record cumulative SNR; best subset = argmax of the curve (random-order baseline alongside)

RQ3 – Subsampling

Subtask · per language – language subset within a family (Belebele)

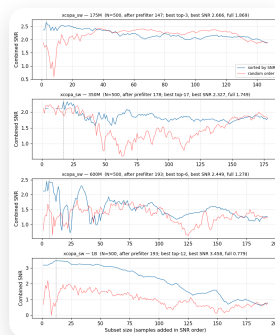
RQ3 – Subsampling

Subtask · per subject – MMLU subject subset curves per size



RQ3 – Subsampling

Per-sample – cumulative SNR over ranked items (xcopa_sw)



RQ3 – Subsampling

Highlighted results

- ✓ **Best subset usually beats the full set substantially** – Global-MMLU
175M +1.52 SNR (`medical_genetics` alone, full 2.12 → 3.65); per-language GMF-tr 1B +1.56 ; Belebele 350M +1.16
- ✓ **Subject subsets beat language subsets** – MMLU subject (mean over 10 langs) gives the most reliable gains
- ✓ **Stability is uneven**: MMLU **subject** picks recur across pools; **language** and **subject × language** picks often flip
- ✓ **Per-item (per-sample) ranking is mostly noise** (cross-size Spearman ≈ 0.05) → tiny argmax (2.5%) subsets overfit and collapse out-of-sample

RQ3 – Subsampling

Results (auto) – top subset gains (SNR: full → best subset)

case	task	size	full → best SNR	+gain
global_mmlu_full_per_language	global_mmlu_full_vi	1B	2.05 → 4.01	+1.95
global_mmlu_full_subjects	global_mmlu_full	175M	2.12 → 3.65	+1.52
per_benchmark	paws	3B	0.37 → 1.81	+1.44
global_mmlu_full_per_language	global_mmlu_full_sw	600M	1.71 → 3.07	+1.36
global_mmlu_full_per_language	global_mmlu_full_vi	350M	1.97 → 3.31	+1.34
global_mmlu_full_per_language	global_mmlu_full_zh	175M	2.15 → 3.46	+1.31
per_benchmark	truthfulqa	3B	0.66 → 1.92	+1.26
per_benchmark	belebele	350M	2.28 → 3.44	+1.16

RQ3 – Subsampling

Proposed improvements to make subset selection trustworthy

- 💡 **Pick a safer subset.** The single best subset is often a fluke. Instead, take the *largest* subset that ties with the peak and still beats the full benchmark. Bigger subsets are more stable.
- 💡 **Average out the noise.** Per-item scores are too noisy to trust. Pool them across seeds and sizes – or pick whole topics, not single items. Coarser units are stabler.
- 💡 **Measure the trust.** Test each pick on a held-out seed, and report how often it survives.

We will do a thorough subsampling analysis on INCLUDE-v2.

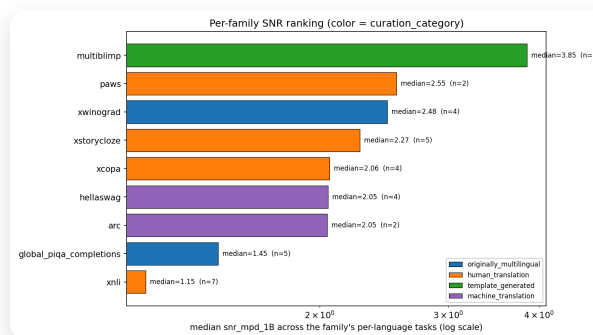
RQ4 – Benchmark Creation

What makes a benchmark high-SNR? What benchmark design features (curation, format, option count, item length) predict SNR?

- ⚙️ **9 above-random benchmark families** (the gate already drops every at-chance 4-option MCQA); SNR signal = per-family median SNR @ 1B from the `custom_swissai_hf` pool
- ⚙️ Per-family aggregate: median across the family's per-language aggregate tasks
- ⚙️ Three phases: **curation** (Phase 0), **task format** (Phase A), **item lengths** (Phase B)
- ⚙️ Statistical tests: Kruskal-Wallis (categorical), Spearman ρ (continuous)
- ⚙️ Length features: 100 items/family sampled from each HF dataset

RQ4 – Benchmark Creation

Per-family median SNR – 9 above-random families (custom_swissai_hf)



RQ4 – Benchmark Creation

Highlighted results

- ✓ **The answer-count penalty is now upstream, in the gate.** It drops every 4-option translated knowledge MCQA (`belebele` , `global_mmlu_full` , `truthfulqa`) before SNR is even computed
- ✓ Top SNR (all 2-option): **multiblimp** (3.9), **paws** (2.5), **xwinograd** (2.5), **xstorycloze** (2.3), **xcopa** (2.1)
- ✓ The only 4-option survivors are `hellaswag` (2.1) and English `arc` (2.0) – contentful; bottom: **global_piqua** (1.5), **xnli** (1.2)
- ✓ **Among the 9 survivors no single design feature is significant** ($n_options$ KW $H = 1.8$, $p = 0.18$; $format$ $H = 0$, $p = 1.0$) – and **curation still explains nothing** ($H = 0.5$, $p = 0.78$)
- ✓ Mechanism: each option adds another noisy log-likelihood estimate to

RQ4 – Benchmark Creation

Results (auto) – per-family SNR, above-random survivors

family	median SNR	n_opts	format
multiblimp	3.85	2	minimal_pair
paws	2.55	2	classification
xwinograd	2.48	2	completion
xstorycloze	2.27	2	completion
xcopa	2.06	2	completion
hellaswag	2.05	4	completion
arc	2.05	4	mcq_question_only
global_pqa_completions	1.45	2	completion
xnli	1.15	3	classification

RQ4 – Benchmark Creation

Proposed methodology improvements

- 💡 **Controlled comparison:** hold format constant, vary curation (HellaSwag MT vs XStoryCloze human translation)
- 💡 Replace marginal KW tests with a **single regression** on (format + n_options + curation)
- 💡 **Add more families** (truthfulqa, mgsm, agieval, ...) – n=9 survivors is underpowered
- 💡 Re-sample **length features from the full multilingual splits**, not just English

Conclusions

Conclusions

Answer to the research question

- **RQ1:** The **dispersion** family (`dist_std`) tracks decision accuracy best (overall $r \approx 0.38$). The *family* ranking holds across seeds (Spearman ρ **+0.80 / +0.93**); the per-language argmax does not.
- **RQ2:** SNR **transfers to AllenAI DataDecide** on the 7 shared English benchmarks – cross-corpus Pearson r **0.92**, Spearman ρ **0.93** (pure 3-seed pool); dispersion + discrepancy families transfer, relative-spread does not.
- **RQ3:** **Subtask subsets beat full benchmarks** (MMLU subjects most stable); per-item selection overfits across scale.

Appendix – Signal & Predictability across Sizes

Appendix – Above-random signal

Custom Apertus pretrains only · mean score per family × size (bold = beats chance + 0.05)

benchmark	rand	175M	350M	600M	1B
multitlimp	0.50	0.90	0.91	0.92	0.93
piqa	0.50	0.67	0.70	0.71	0.73
xwinograd	0.50	0.60	0.64	0.67	0.70
xstorycloze	0.50	0.54	0.55	0.57	0.58
xcopa	0.50	0.54	0.55	0.56	0.56
global_piqa_completions	0.50	0.50	0.52	0.53	0.54
paws	0.50	0.51	0.51	0.51	0.52
xnli	0.33	0.38	0.39	0.40	0.40
hellaswag	0.25	0.29	0.31	0.33	0.34
arc	0.25	0.26	0.27	0.29	0.30
truthfulqa	0.25	0.26	0.26	0.26	0.26
agieval_sat	0.25	0.26	0.25	0.26	0.24
mmlu	0.25	0.25	0.25	0.25	0.25
belebele	0.25	0.24	0.25	0.25	0.25
global_mmlu_full	0.25	0.25	0.25	0.25	0.25
global_mmlu	0.25		0.25		
arabic_leaderboard_alghafa_mcq_exams_test	0.25	0.24	0.25	0.24	0.24
agieval_logiqa	0.25	0.22	0.22	0.23	0.23
truthfulqa_mc1	0.25	0.23	0.22	0.22	0.23
openbookqa	0.25	0.20	0.22	0.23	0.25
agieval_lsats	0.20	0.22	0.22	0.22	0.22
commonsense_qa	0.20	0.20	0.21	0.21	0.20
agieval	0.25	0.18	0.18	0.18	0.18

Appendix – Above-random signal

All models (custom + Swiss-AI/HF refs) · mean score per family × size (bold = beats chance + 0.05)

benchmark	rand	175M	270M	350M	600M	1B	1.7B	3B	4B	7-9B	12-14B	27-32B	70B
multitlimp	0.50	0.90	0.91	0.91	0.92	0.93	0.94	0.92	0.96	0.91	0.94	0.94	0.98
piqa	0.50	0.67	0.68	0.70	0.71	0.73	0.76	0.77	0.79	0.79	0.81	0.81	0.83
xwinograd	0.50	0.60	0.63	0.64	0.67	0.70	0.72	0.76	0.77	0.76	0.79	0.79	0.82
xstorycloze	0.50	0.54	0.54	0.55	0.56	0.58	0.59	0.62	0.64	0.63	0.67	0.68	0.71
xcopa	0.50	0.54	0.56	0.55	0.56	0.57	0.59	0.60	0.64	0.60	0.65	0.64	0.70
global_piqa_completions	0.50	0.50	0.51	0.52	0.53	0.55	0.57	0.61	0.63	0.60	0.65	0.65	0.74
paws	0.50	0.51	0.52	0.51	0.51	0.53	0.58	0.57	0.60	0.57	0.60	0.61	0.60
agieval_sat	0.25	0.26	0.30	0.25	0.29	0.25	0.64	0.41	0.78	0.65	0.83	0.83	0.74
commonsense_qa	0.20	0.20	0.21	0.21	0.24	0.22	0.73	0.48	0.70	0.69	0.77	0.76	0.54
mmlu	0.25	0.25	0.26	0.25	0.27	0.27	0.61	0.46	0.65	0.63	0.71	0.73	0.65
belebele	0.25	0.24	0.24	0.25	0.27	0.26	0.60	0.43	0.66	0.55	0.70	0.69	0.68
xnli	0.33	0.38	0.39	0.39	0.40	0.41	0.42	0.42	0.44	0.41	0.43	0.43	0.46
global_mmlu_full	0.25	0.25	0.26	0.25	0.26	0.25	0.46	0.38	0.52	0.46	0.56	0.56	0.53
hellaswag	0.25	0.29	0.29	0.31	0.33	0.34	0.36	0.40	0.41	0.40	0.44	0.45	0.49
global_mmlu	0.25			0.25				0.49					
arc	0.25	0.26	0.25	0.27	0.29	0.30	0.36	0.37	0.43	0.40	0.46	0.46	0.47
truthfulqa	0.25	0.26	0.35	0.26	0.35	0.35	0.38	0.35	0.38	0.38	0.38	0.37	0.40
arabic_leaderboard_alghafa_mcq_exams_test	0.25	0.24	0.25	0.25	0.24	0.24	0.37	0.36	0.43	0.36	0.45	0.45	0.48
agieval_logiqa	0.25	0.22	0.21	0.22	0.23	0.23	0.32	0.26	0.39	0.35	0.42	0.39	0.36
agieval	0.25	0.18	0.18	0.18	0.19	0.18	0.35	0.25	0.41	0.36	0.46	0.46	0.36
openbookqa	0.25	0.20	0.21	0.22	0.24	0.26	0.30	0.30	0.33	0.34	0.36	0.36	0.38
truthfulqa_mc1	0.25	0.23	0.24	0.22	0.23	0.23	0.32	0.27	0.32	0.31	0.35	0.34	0.36
agieval_lsats	0.20	0.22	0.24	0.22	0.21	0.21	0.25	0.23	0.21	0.22	0.25	0.24	0.21

Appendix – Decision accuracy across sizes

English (en) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
hellaswag	0.92	0.86	0.83	0.89	0.86	0.80	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.75	0.83	0.81	0.81	0.83	0.76	1.00	1.00	1.00	1.00	1.00	1.00
openbookqa	0.78	0.75	0.67	0.81	0.72	0.84	1.00	1.00	0.00	1.00	1.00	1.00
piqa	0.86	0.89	0.89	0.86	0.86	0.76	1.00	0.00	1.00	1.00	1.00	0.00
xwinograd	0.86	0.81	0.72	0.78	0.81	0.73	0.00	0.00	1.00	1.00	1.00	1.00
commonsense_qa	0.58	0.42	0.36	0.33	0.44	0.53	1.00	1.00	1.00	1.00	1.00	1.00
agieval_logiqa	0.56	0.61	0.67	0.44	0.56	0.44	1.00	1.00	1.00	1.00	0.00	1.00
arc_easy	0.78	0.81	0.89	0.86	0.83	0.78	0.00	0.00	1.00	0.00	1.00	1.00
arc_challenge	0.69	0.72	0.83	0.92	0.86	0.80	0.00	0.00	1.00	0.00	1.00	1.00
belebele	0.64	0.69	0.75	0.72	0.78	0.78	0.00	0.00	1.00	1.00	0.00	1.00
paws	0.50	0.47	0.42	0.58	0.53	0.78	1.00	1.00	0.00	0.00	1.00	1.00
xnli	0.64	0.53	0.33	0.78	0.42	0.51	1.00	1.00	0.00	0.00	1.00	1.00
agieval_sat	0.31	0.36	0.42	0.61	0.50	0.73	0.00	0.00	1.00	1.00	1.00	1.00
global_mmlu_full	0.39	0.33	0.64	0.44	0.36	0.58	0.00	0.00	1.00	1.00	1.00	1.00
truthfulqa_mc1	0.33	0.53	0.33	0.36	0.78	0.33	0.00	1.00	1.00	1.00	1.00	0.00
mmlu	0.50	0.06	0.58	0.50	0.47	0.42	0.00	0.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.61	0.50	0.67	0.72	0.28	0.40	0.00	0.00	1.00	0.00	1.00	1.00
multiblimp	0.61	0.50	0.78	0.83	0.61	0.71	1.00	0.00	0.00	1.00	0.00	0.00
agieval	0.25	0.44	0.61	0.36	0.47	0.53	0.00	0.00	1.00	1.00	0.00	1.00

Appendix – Decision accuracy across sizes

Arabic (ar) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
hellaswag	0.78	0.78	0.72	0.83	0.89	0.82	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.75	0.89	0.83	0.69	0.75	0.78	1.00	1.00	1.00	1.00	1.00	1.00
multitlimp	0.69	0.83	0.83	0.86	0.75	0.89	1.00	1.00	1.00	1.00	0.00	1.00
truthfulqa_ar_mc1	0.86	0.58	0.50	0.61	0.58	0.67	1.00	1.00	1.00	1.00	1.00	1.00
belebele	0.69	0.42	0.64	0.61	0.56	0.53	1.00	1.00	1.00	1.00	0.00	1.00
truthfulqa_ar_mc2							0.00	0.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.58	0.39	0.44	0.47	0.53	0.49	1.00	1.00	1.00	1.00	0.00	1.00
xnli	0.69	0.17	0.42	0.36	0.50	0.62	1.00	1.00	1.00	1.00	0.00	1.00
arabic_leaderboard_alghafa_mcq_exams_test	0.33	0.61	0.36	0.44	0.69	0.31	1.00	1.00	1.00	1.00	0.00	1.00
global_mmlu_full	0.47	0.61	0.50	0.53	0.53	0.80	0.00	0.00	1.00	1.00	1.00	1.00
arc	0.31	0.72	0.69	0.42	0.44	0.64	1.00	1.00	0.00	1.00	0.00	1.00
agieval_lsats	0.61	0.36	0.50	0.53	0.61	0.64	0.00	1.00	1.00	1.00	0.00	0.00

Appendix – Decision accuracy across sizes

Spanish (es) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
hellaswag	0.97	0.81	0.86	0.83	0.83	0.82	1.00	1.00	1.00	1.00	1.00	1.00
multiblimp	0.78	0.81	0.89	0.64	0.67	0.82	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.58	0.75	0.72	0.72	0.86	0.71	1.00	1.00	1.00	1.00	1.00	1.00
global_piqqa_completions	0.81	0.64	0.67	0.61	0.69	0.38	1.00	1.00	1.00	1.00	1.00	1.00
arc	0.53	0.67	0.67	0.47	0.47	0.58	1.00	1.00	1.00	1.00	0.00	1.00
paws	0.56	0.33	0.42	0.56	0.53	0.60	1.00	1.00	0.00	1.00	1.00	1.00
global_mmlu_full	0.50	0.36	0.47	0.47	0.64	0.73	0.00	0.00	1.00	1.00	1.00	1.00
xnli	0.39	0.47	0.44	0.64	0.61	0.51	1.00	1.00	1.00	0.00	0.00	1.00
truthfulqa_es_mc1	0.75	0.42	0.53	0.28	0.56	0.53	0.00	0.00	1.00	1.00	1.00	1.00
belebele	0.50	0.22	0.53	0.56	0.36	0.62	0.00	0.00	1.00	1.00	1.00	1.00
truthfulqa_es_mc2							1.00	0.00	1.00	1.00	0.00	0.00

Appendix – Decision accuracy across sizes

Basque (eu) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
truthfulqa_eu_mc2							1.00	1.00	1.00	1.00	0.00	1.00
xstorycloze	0.92	0.69	0.75	0.72	0.72	0.82	1.00	1.00	1.00	1.00	0.00	1.00
hellaswag	0.75	0.56	0.61	0.47	0.58	0.58	1.00	1.00	1.00	1.00	1.00	1.00
multitlimp	0.64	0.44	0.69	0.53	0.61	0.60	1.00	1.00	1.00	1.00	1.00	1.00
arc	0.69	0.75	0.72	0.78	0.69	0.76	1.00	1.00	1.00	1.00	0.00	1.00
xnli	0.47	0.44	0.56	0.75	0.64	0.76	1.00	1.00	1.00	1.00	0.00	1.00
paws	0.39	0.50	0.36	0.67	0.58	0.56	1.00	1.00	0.00	0.00	1.00	1.00
belebele	0.50	0.56	0.53	0.56	0.47	0.67	0.00	0.00	1.00	1.00	0.00	1.00
truthfulqa_eu_mc1	0.36	0.44	0.56	0.42	0.64	0.60	0.00	0.00	1.00	1.00	0.00	1.00
xcopa	0.39	0.44	0.44	0.50	0.50	0.76	0.00	0.00	1.00	0.00	0.00	1.00

Appendix – Decision accuracy across sizes

Hindi (hi) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
multiblimp	0.89	0.86	0.83	0.86	0.78	0.89	1.00	1.00	1.00	1.00	1.00	1.00
hellaswag	0.75	0.81	0.78	0.83	0.81	0.89	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.61	0.58	0.64	0.64	0.81	0.71	1.00	1.00	1.00	1.00	1.00	1.00
global_mmlu_full	0.39	0.53	0.53	0.64	0.53	0.38	1.00	1.00	1.00	1.00	1.00	1.00
arc	0.33	0.58	0.31	0.42	0.58	0.67	1.00	1.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.39	0.53	0.42	0.36	0.36	0.76	1.00	1.00	1.00	1.00	1.00	1.00
belebele	0.53	0.39	0.61	0.69	0.53	0.51	1.00	1.00	1.00	1.00	0.00	1.00
truthfulqa_hi_mc2							0.00	0.00	1.00	1.00	1.00	1.00
xnli	0.78	0.56	0.72	0.56	0.72	0.64	1.00	1.00	0.00	0.00	1.00	1.00
truthfulqa_hi_mc1	0.58	0.67	0.47	0.53	0.61	0.49	0.00	0.00	1.00	1.00	1.00	1.00

Appendix – Decision accuracy across sizes

Japanese (ja) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
xwinograd	0.61	0.78	0.67	0.61	0.67	0.56	1.00	1.00	1.00	1.00	1.00	1.00
belebele	0.42	0.72	0.64	0.53	0.56	0.84	1.00	1.00	1.00	1.00	1.00	1.00
paws	0.44	0.89	0.56	0.50	0.56	0.64	1.00	1.00	1.00	1.00	1.00	1.00
global_mmlu_full	0.61	0.58	0.58	0.53	0.53	0.53	1.00	1.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.56	0.58	0.56	0.69	0.78	0.62	1.00	1.00	1.00	1.00	0.00	1.00

Appendix – Decision accuracy across sizes

Russian (ru) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
truthfulqa_ru_mc2							1.00	1.00	1.00	1.00	1.00	1.00
hellaswag	0.92	0.92	0.92	0.89	0.89	0.73	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.72	0.81	0.81	0.86	0.81	0.84	1.00	1.00	1.00	1.00	1.00	1.00
multiblimp	0.89	0.89	0.81	0.89	0.86	0.82	1.00	1.00	1.00	1.00	0.00	1.00
global_piqa_completions	0.81	0.72	0.75	0.86	0.83	0.84	1.00	1.00	0.00	1.00	1.00	1.00
belebele	0.61	0.47	0.75	0.58	0.64	0.47	1.00	1.00	1.00	1.00	0.00	1.00
xwinograd	0.61	0.69	0.53	0.58	0.58	0.42	1.00	1.00	0.00	1.00	1.00	1.00
xnli	0.47	0.50	0.25	0.36	0.44	0.64	1.00	1.00	1.00	0.00	1.00	1.00
global_mmlu_full	0.39	0.56	0.69	0.56	0.47	0.44	0.00	0.00	1.00	1.00	1.00	1.00
arc	0.33	0.58	0.44	0.47	0.72	0.49	1.00	1.00	0.00	1.00	0.00	1.00
truthfulqa_ru_mc1	0.33	0.47	0.31	0.69	0.47	0.36	0.00	1.00	1.00	1.00	1.00	0.00

Appendix – Decision accuracy across sizes

Swahili (sw) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
xcopa	0.56	0.64	0.78	0.42	0.67	0.56	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.58	0.72	0.67	0.53	0.75	0.69	1.00	1.00	1.00	1.00	0.00	1.00
xnli	0.39	0.39	0.33	0.50	0.67	0.64	1.00	1.00	1.00	1.00	1.00	1.00
global_mmlu_full	0.47	0.47	0.56	0.78	0.58	0.49	0.00	0.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.39	0.50	0.72	0.44	0.50	0.40	1.00	1.00	1.00	0.00	0.00	1.00
belebele	0.47	0.50	0.64	0.69	0.44	0.51	0.00	0.00	1.00	1.00	0.00	1.00

Appendix – Decision accuracy across sizes

Thai (th) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
xnli	0.72	0.67	0.81	0.56	0.69	0.69	1.00	1.00	1.00	1.00	1.00	1.00
belebele	0.61	0.67	0.64	0.72	0.64	0.78	0.00	0.00	1.00	1.00	1.00	1.00
xcopa	0.31	0.64	0.69	0.39	0.11	0.53	1.00	1.00	0.00	1.00	1.00	1.00
global_piqa_completions	0.56	0.67	0.47	0.44	0.42	0.67	1.00	1.00	0.00	1.00	0.00	1.00

Appendix – Decision accuracy across sizes

Turkish (tr) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
multiblimp	0.89	0.69	0.89	0.75	0.94	0.80	1.00	1.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.78	0.69	0.64	0.64	0.64	0.69	1.00	1.00	1.00	1.00	1.00	1.00
global_mmlu_full	0.56	0.39	0.64	0.50	0.42	0.44	1.00	1.00	1.00	1.00	1.00	1.00
belebele	0.86	0.47	0.61	0.61	0.58	0.62	1.00	1.00	1.00	1.00	0.00	1.00
xnli	0.72	0.61	0.64	0.72	0.53	0.44	1.00	1.00	1.00	1.00	0.00	1.00
xcopa	0.42	0.53	0.25	0.33	0.50	0.56	1.00	1.00	1.00	1.00	1.00	1.00

Appendix – Decision accuracy across sizes

Vietnamese (vi) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
hellaswag	0.83	0.81	0.92	0.86	0.75	0.78	1.00	1.00	1.00	1.00	1.00	1.00
truthfulqa_vi_mc1	0.72	0.86	0.33	0.58	0.56	0.24	1.00	1.00	1.00	1.00	1.00	1.00
arc	0.64	0.69	0.53	0.72	0.56	0.71	1.00	1.00	0.00	1.00	1.00	1.00
xnli	0.39	0.61	0.53	0.61	0.64	0.71	1.00	1.00	1.00	0.00	1.00	1.00
global_pqa_completions	0.53	0.47	0.75	0.50	0.50	0.62	1.00	1.00	1.00	1.00	0.00	1.00
xcopa	0.53	0.69	0.67	0.78	0.81	0.84	1.00	1.00	0.00	1.00	0.00	1.00
global_mmlu_full	0.86	0.58	0.72	0.50	0.58	0.40	0.00	0.00	1.00	1.00	1.00	1.00
belebele	0.58	0.56	0.50	0.47	0.42	0.64	0.00	0.00	1.00	1.00	0.00	1.00
truthfulqa_vi_mc2							0.00	1.00	1.00	1.00	0.00	0.00

Appendix – Decision accuracy across sizes

Chinese (zh) · small→large size pair (bold ≥ 0.75)

benchmark	175M→350M	175M→600M	175M→1B	350M→600M	350M→1B	600M→1B	1B→12-14B	1B→27-32B	4B→12-14B	7-9B→12-14B	7-9B→27-32B	12-14B→27-32B
truthfulqa_zh_mc2							1.00	1.00	1.00	1.00	0.00	1.00
agieval_logiqa	0.47	0.64	0.39	0.44	0.58	0.67	1.00	1.00	1.00	1.00	1.00	1.00
global_piqa_completions	0.67	0.67	0.39	0.50	0.56	0.40	1.00	1.00	1.00	1.00	1.00	1.00
xwinograd	0.36	0.56	0.42	0.64	0.50	0.36	1.00	1.00	1.00	1.00	1.00	1.00
xstorycloze	0.44	0.50	0.67	0.61	0.61	0.71	1.00	1.00	1.00	1.00	0.00	1.00
xcopa	0.50	0.33	0.42	0.67	0.64	0.67	1.00	1.00	1.00	1.00	0.00	1.00
arc	0.44	0.47	0.39	0.42	0.50	0.60	1.00	1.00	0.00	1.00	1.00	1.00
paws	0.39	0.56	0.58	0.44	0.58	0.51	1.00	0.00	1.00	1.00	1.00	0.00
global_mmlu_full	0.36	0.56	0.64	0.42	0.28	0.51	0.00	0.00	1.00	1.00	1.00	1.00
truthfulqa_zh_mc1	0.28	0.58	0.53	0.42	0.36	0.47	0.00	1.00	1.00	1.00	1.00	0.00
belebele	0.67	0.25	0.50	0.58	0.72	0.60	0.00	0.00	1.00	1.00	0.00	1.00
xnli	0.47	0.58	0.67	0.44	0.47	0.67	1.00	1.00	0.00	0.00	0.00	1.00